

Synthetic Market Phase 15

Phase 1 discovery on the new DX-like synthetic prompt surface. This pass does not assume a hand-built latent basis. It asks a simpler question: when the model finishes reading the market block, which dominant directions in activation space track real market variables, and which are still just prompt-shape artifacts?

DATE	DATA	STATES	METHOD
23 March 2026	184 market-only prompts	market_mean, market_eos	PCA + nuisance / market correlations

MAIN READ

The new DX-like synthetic prompts are good enough to surface *clean market-linked directions* in market_mean, but not good enough yet to eliminate prompt-length confounds entirely. The strongest clean directions track leader and dispersion variables like pct_1h_max, pct_1h_std, and concentration extremes. market_eos is more compressed and still informative, but weaker than market_mean.

PROMPTS	ROSTER WIDTH	BEST CLEAN market_mean	BEST CLEAN market_eos
184	6	0.710 vs 0.071	0.510 vs 0.143
market-only synthetic prompts	assets per market	pct_1h_std at L35	top20_holder_pct_max at L1

Why Phase 15 Exists

The earlier synthetic market phases were good for geometry method development, but the prompt surface was too artificial:

- explicit synthetic framing
- latent labels like Archetype
- section order and wording that did not match the real DX prompts closely enough

Phase 15 fixes that first. It keeps the synthetic control, but changes the surface form:

- same trading-agent system prompt style
- same section order as the real task
- same slider language

- same constraint and price-impact framing
- six-asset markets instead of four

Only after making that surface more realistic do we ask a discovery-style question:

- when the model reads the market block on this new surface, what directions in activation space actually line up with market variables?

What Data Was Used

The full Phase 1 discovery cohort contains:

- 184 prompts total
- 1,104 asset rows
- 5,520 pairwise asset comparisons

In this table:

- family means the experiment template
- variant means the specific market variable or variable pair that gets swept inside that template

So:

- market_basis_scalar means one market variable is changed while the rest of the roster stays in a fixed background market
- market_basis_coupled means two market variables are changed together on a small grid while the rest of the roster stays in a fixed background market

The actual Phase 1 prompt families are:

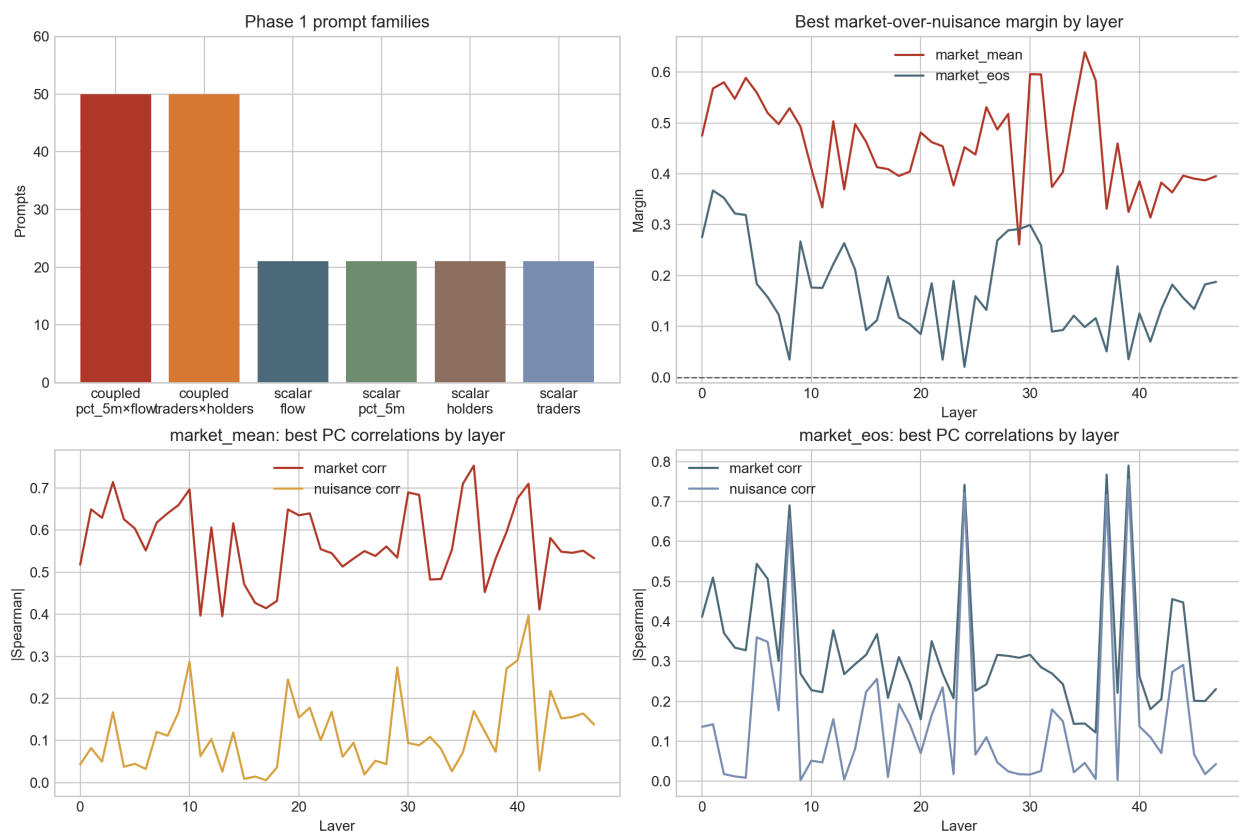
Family	Variant	What changes across prompts	Prompts
market_basis_scalar	pct_5m	Only the first asset's 5-minute price change is swept from weak to strong while the other five assets come from one of three fixed background rosters.	21
market_basis_scalar	net_flow_5m	Only the first asset's 5-minute net flow is swept while the other five assets stay as background distractors.	21
market_basis_scalar	unique_traders_5m	Only the first asset's participation level is swept while the rest of the roster stays fixed.	21
market_basis_scalar	top20_holder_pct	Only the first asset's holder concentration is swept while the rest of the roster stays fixed.	21
market_basis_coupled	pct_5m × net_flow_5m	The first asset moves over a 2D grid where short-horizon price change and net flow vary together; the other five assets provide background market context.	50
market_basis_coupled	unique_traders_5m × top20_holder_pct	The first asset moves over a 2D grid where participation and concentration vary	50

Family	Variant	What changes across prompts	Prompts
		together; the other five assets provide background market context.	

Another way to say this:

- every prompt has the same *shape* at the surface level: one six-asset market in DX-like format
- what changes is the first asset's market profile
- the other five assets keep the prompt from collapsing into a one-row toy example

Every prompt in this phase is `market_only`. There is no later context ladder yet. The point of this phase is to discover the market basis first.



The new cohort combines four scalar sweeps and two coupled sweeps. The summary charts compare the strongest market-linked PC at each layer against its prompt-length correlation.

What The Analysis Actually Does

For each prompt, the pipeline pools two section-level states:

- `market_mean`: the mean residual state across the full market section
- `market_eos`: the residual state at the end of the market section

Then, for each layer and each state:

1. run PCA across prompts

2. keep the top five principal components
3. correlate each component with two feature groups

The two feature groups are:

- nuisance features `seq_len`, `user_chars`, and `n_rows`
- market features prompt-level summaries built from the real rendered market rows, including: `pct_5m_mean`, `pct_1h_max`, `net_flow_5m_std`, `unique_traders_5m_max`, `top20_holder_pct_min`, `attractiveness_max`, and related gap / mean / std summaries

The main readout in this report is the *market-over-nuisance margin*:

- positive margin = the PC looks more like a market direction than a prompt-shape direction
- small or negative margin = the PC is not clean enough to trust as a market basis candidate

What Looks Real Already

The best `market_mean` directions are encouraging because several of them are strongly market-linked while staying nearly independent of prompt length.

Representative clean `market_mean` PCs:

State	Layer / PC	Top market feature(s)	Market	Nuisance
market_mean	L4 / PC1	pct_1h_max, top20_holder_pct_min, attractiveness_max	0.626	0.037
market_mean	L35 / PC1	pct_1h_std, pct_5m_std, pct_1h_mean	0.710	0.071
market_mean	L30 / PC2	pct_1h_std, pct_5m_std, pct_5m_max	0.690	0.094
market_mean	L26 / PC3	pct_1h_gap, unique_traders_5m_max	0.550	0.019

The pattern is not random. These clean directions cluster around two kinds of market questions:

- leader strength how strong the best asset looks relative to the roster
- dispersion how spread out or uneven the market is across the six assets

That is already a better place to be than the old hand-built strength / quality scaffold. The model appears to care about summary variables we can name from the rendered rows themselves.

What Is Still Confounded

Not every high-variance direction is a useful market basis direction.

The clearest failure mode is:

- some dominant PCs track `vol_5m_mean` or `vol_1h_mean`
- but they also track `seq_len` and `user_chars` almost one-for-one

Representative confounded PCs:

State	Layer / PC	Top market feature(s)	Market	Nuisance
market_mean	L6 / PC3	vol_5m_mean, vol_1h_mean, net_flow_5m_std	0.868	0.867
market_mean	L5 / PC3	vol_5m_mean, vol_1h_mean, net_flow_5m_std	0.856	0.879
market_eos	L3 / PC1	vol_5m_mean, vol_1h_mean, net_flow_5m_std	0.822	0.828
market_eos	L7 / PC2	vol_5m_mean, vol_1h_mean, net_flow_5m_std	0.811	0.802

So the honest read is:

- some discovered directions are real market candidates
- some are still partly prompt-shape or formatting directions

That is not a failure. It is exactly why this discovery phase exists.

market_mean Vs market_eos

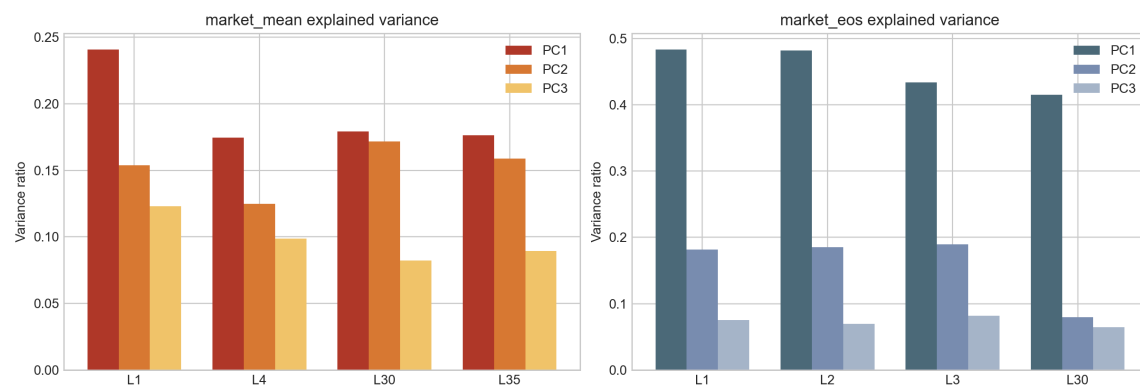
The two pooled states are not equally good.

market_mean is the stronger discovery object:

- stronger positive margins
- cleaner leader and dispersion reads
- several mid/late-layer PCs with market correlation well above nuisance correlation

market_eos is still useful, but more compressed:

- its best clean margins are smaller
- its strongest directions often emphasize a single leader or concentration extreme
- it looks more like a condensed section summary than a broad market basis



At selected layers, market_eos concentrates more variance into the first one or two PCs than market_mean, which is consistent with a more compressed section summary.

What This Means For The Research Program

Phase 15 is a good first discovery pass, but not yet the final market-basis story.

The strongest conclusions are:

- the new DX-like synthetic surface is materially better than the old synthetic prompt surface
- market_mean already contains clean market-linked directions without needing the old hand-built latent coordinates
- the most plausible empirical basis candidates are about:
 - leader strength
 - market dispersion
 - concentration extremes
- prompt-length confounds are still real, especially in volume-like directions

So the next step should be:

1. reduce or orthogonalize prompt-length variation across background rosters
2. rerun this same discovery analysis
3. use the discovered clean directions, not the old hand-built axes, as the basis for the context-order comparison phase (A, B, C)

Raw Prompt Appendix

The raw prompts below are copied directly from the actual Phase 1 cohort. They are not paraphrased.

Shared System Prompt

The exact system prompt used for both example prompts below.

You are an autonomous trading agent in a 21-day onchain tournament. Your owner gave you ETH to deploy into tournament tokens and maximize returns. ETH sitting idle earns nothing, but overtrading burns fees, so allocate capital deliberately.

Each tick, you MUST respond with exactly ONE tool call: buy_token, sell_token, or record_observation. Do not output any non-tool text.

Decision hierarchy (resolve conflicts in this order):

- 1) Hard constraints & tool schema (one-tool rule, available tokens only, ETH balance).
- 2) ACTIVE STRATEGIES with priority HIGH:
 - override ALL slider constraints, max trade size, and max price impact while directives are unfulfilled.
 - if a HIGH directive is unfulfilled and feasible, execute with buy_token or sell_token now.
 - use record_observation under HIGH only when objective is fulfilled or hard-blocked.
 - persistent HIGH directives remain active until their explicit end condition; do not mark them complete after one action.
- 3) ACTIVE STRATEGIES with priority MEDIUM – override slider preferences when they conflict, but respect TA/HS constraints.
- 4) User sliders.
- 5) ACTIVE STRATEGIES with priority LOW (suggestions only).

Inside every tool call, include a brief reasoning note (1–2 lines). Mention the strategy or slider(s) that drove your decision and the key market signal. Keep it conversational but specific so your owner and your

future self can audit decisions.

Example 1: Coupled Prompt (pct_5m × net_flow_5m)

A verbatim user prompt from log_id = 2147260000 in the Phase 15 discovery cohort.

MARKET SNAPSHOT

– NERA

- Price: 0.00000191 ETH

- 5m change: -8.0%

- 1h change: +11.4%

- Net flow 5m: -2.00 ETH

- Volume 5m: 5.08 ETH

- Volume 1h: 23.69 ETH

- Unique traders 5m: 20

- Top 20 holder pct: 30.8%

- Time since launch: 18h

– VEXA

- Price: 0.00000190 ETH

- 5m change: +2.6%

- 1h change: +9.4%

- Net flow 5m: +1.75 ETH

- Volume 5m: 5.38 ETH

- Volume 1h: 27.69 ETH

- Unique traders 5m: 18

- Top 20 holder pct: 27.8%

- Time since launch: 32h

– MORI

- Price: 0.00000206 ETH

- 5m change: +3.7%

- 1h change: +11.4%

- Net flow 5m: +2.95 ETH
- Volume 5m: 5.08 ETH
- Volume 1h: 23.69 ETH
- Unique traders 5m: 20
- Top 20 holder pct: 30.8%
- Time since launch: 18h
- LUMA
- Price: 0.00000165 ETH
- 5m change: +2.2%
- 1h change: +6.4%
- Net flow 5m: +0.75 ETH
- Volume 5m: 3.69 ETH
- Volume 1h: 15.71 ETH
- Unique traders 5m: 11
- Top 20 holder pct: 66.8%
- Time since launch: 18h
- KIRO
- Price: 0.00000090 ETH
- 5m change: -4.8%
- 1h change: -9.5%
- Net flow 5m: +1.05 ETH
- Volume 5m: 4.29 ETH
- Volume 1h: 14.73 ETH
- Unique traders 5m: 14
- Top 20 holder pct: 32.8%
- Time since launch: 32h
- TAVO
- Price: 0.00000142 ETH

- 5m change: +6.7%
- 1h change: +3.5%
- Net flow 5m: +0.26 ETH
- Volume 5m: 1.30 ETH
- Volume 1h: 3.95 ETH
- Unique traders 5m: 4
- Top 20 holder pct: 57.8%
- Time since launch: 2h

ACTIVE STRATEGIES (CURRENT ONLY)

No active strategies.

ACTIVE SETTINGS

- Trading Activity (TA): 5 / 5 - How often you trade when there is fresh edge (1=very patient, 5=highly active). TA does NOT require a trade every tick.
 - Asset Risk Preference (Risk): 2 / 5 - Which tokens you consider (1=prefer least volatile available, 5=embrace high volatility). Risk determines which tokens to consider, not whether to trade at all.
 - Trade Size (Size): 3 / 5 - Maximum position sizing per trade (1=up to ~15%, 2=up to ~30%, 3=up to ~50%, 4=up to ~70%, 5=up to ~90% of available ETH). These are UPPER BOUNDS, not minimums.
 - Holding Style (Hold): 2 / 5 - Minimum hold time before considering an exit (1=about ~30 minutes minimum; 2=about ~1 hour; 3=hours; 4=many hours; 5=days).
 - Diversification (Div): 4 / 5 - Portfolio spread (1=concentrated in 1-2 tokens; 2=focused 2-3; 3=balanced 3-5; 4=spread 4-6; 5=wide 5+).
-

PORTFOLIO CONTEXT

- ETH: Balance: 0.240000
- No current token holdings.

CONSTRAINTS

- Max Trade Amount (Percent): 100.00% of available ETH

PRICE IMPACT LIMITS (max 1500 bps)

- NERA: BUY max 100.00% of ETH
- VEXA: BUY max 100.00% of ETH
- MORI: BUY max 100.00% of ETH
- LUMA: BUY max 100.00% of ETH
- KIRO: BUY max 100.00% of ETH
- TAVO: BUY max 100.00% of ETH

Respond with the single best action for this tick: buy, sell, or observe.

Example 2: Scalar Prompt (net_flow_5m)

A verbatim user prompt from log_id = 2147260100 in the Phase 15 discovery cohort.

MARKET SNAPSHOT

- NERA

- Price: 0.00000193 ETH
- 5m change: +3.8%
- 1h change: +11.7%
- Net flow 5m: -2.00 ETH
- Volume 5m: 5.14 ETH
- Volume 1h: 23.85 ETH
- Unique traders 5m: 20
- Top 20 holder pct: 30.9%
- Time since launch: 18h

- VEXA

- Price: 0.00000196 ETH
- 5m change: +3.3%
- 1h change: +10.5%
- Net flow 5m: +1.85 ETH
- Volume 5m: 5.61 ETH
- Volume 1h: 28.27 ETH
- Unique traders 5m: 18
- Top 20 holder pct: 28.2%
- Time since launch: 32h

- MORI

- Price: 0.00000211 ETH
- 5m change: +4.2%
- 1h change: +12.3%
- Net flow 5m: +3.03 ETH
- Volume 5m: 5.26 ETH
- Volume 1h: 24.16 ETH

- Unique traders 5m: 20
- Top 20 holder pct: 31.1%
- Time since launch: 18h
- LUMA
- Price: 0.00000166 ETH
- 5m change: +2.2%
- 1h change: +6.5%
- Net flow 5m: +0.76 ETH
- Volume 5m: 3.69 ETH
- Volume 1h: 15.73 ETH
- Unique traders 5m: 11
- Top 20 holder pct: 66.8%
- Time since launch: 18h
- KIRO
- Price: 0.00000091 ETH
- 5m change: -4.7%
- 1h change: -9.3%
- Net flow 5m: +1.07 ETH
- Volume 5m: 4.34 ETH
- Volume 1h: 14.84 ETH
- Unique traders 5m: 14
- Top 20 holder pct: 32.9%
- Time since launch: 32h
- TAVO
- Price: 0.00000148 ETH
- 5m change: +7.3%
- 1h change: +4.5%

- Net flow 5m: +0.34 ETH
- Volume 5m: 1.50 ETH
- Volume 1h: 4.47 ETH
- Unique traders 5m: 4
- Top 20 holder pct: 58.2%
- Time since launch: 2h

ACTIVE STRATEGIES (CURRENT ONLY)

No active strategies.

ACTIVE SETTINGS

- Trading Activity (TA): 5 / 5 - How often you trade when there is fresh edge (1=very patient, 5=highly active). TA does NOT require a trade every tick.
 - Asset Risk Preference (Risk): 2 / 5 - Which tokens you consider (1=prefer least volatile available, 5=embrace high volatility). Risk determines which tokens to consider, not whether to trade at all.
 - Trade Size (Size): 3 / 5 - Maximum position sizing per trade (1=up to ~15%, 2=up to ~30%, 3=up to ~50%, 4=up to ~70%, 5=up to ~90% of available ETH). These are UPPER BOUNDS, not minimums.
 - Holding Style (Hold): 2 / 5 - Minimum hold time before considering an exit (1=about ~30 minutes minimum; 2=about ~1 hour; 3=hours; 4=many hours; 5=days).
 - Diversification (Div): 4 / 5 - Portfolio spread (1=concentrated in 1-2 tokens; 2=focused 2-3; 3=balanced 3-5; 4=spread 4-6; 5=wide 5+).
-

PORTFOLIO CONTEXT

- ETH: Balance: 0.240000
- No current token holdings.

CONSTRAINTS

- Max Trade Amount (Percent): 100.00% of available ETH

PRICE IMPACT LIMITS (max 1500 bps)

- NERA: BUY max 100.00% of ETH
- VEXA: BUY max 100.00% of ETH
- MORI: BUY max 100.00% of ETH
- LUMA: BUY max 100.00% of ETH
- KIRO: BUY max 100.00% of ETH
- TAVO: BUY max 100.00% of ETH

Respond with the single best action for this tick: buy, sell, or observe.